

HIBA MESSAI

Data Engineer | Geospatial Data Platforms & Applied AI

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PROFESSIONAL SUMMARY

Final-year engineering student (ENSTA-B, class of 2027) in data platforms and AI. Built a containerized geospatial data platform at a Canadian startup (Docker Compose, PostgreSQL, FastAPI; 37 open datasets, SCD Type 2). Seeking a final-year (PFE) internship in data engineering, ideally geospatial or energy.

EDUCATION

Engineering Degree in Advanced Technologies, AI & Energy Systems 2024 - 2027
ENSTA-B, Tunisia
Preparatory Cycle in Mathematics & Physics, ISSAT Mahdia (2022 - 2024) | Baccalaureate in Mathematics (2021)

TECHNICAL SKILLS

Data Engineering: ETL/ELT Pipelines, PostgreSQL, SQL, Data Warehousing (SCD Type 2), Schema Validation, Data Quality, SLA Monitoring

Backend & DevOps: Python, FastAPI, REST APIs, Docker, Docker Compose, Git/GitHub, Pytest, Pandas, NumPy, Streamlit, Plotly; also SQL, C, Java, MATLAB, R

Geospatial: Spatial Analysis, Heatmaps, SciPy cKDTree, Dijkstra Routing

ML & GenAI: Time-Series Forecasting (XGBoost, Random Forest, LSTM), CNNs, Autoencoders, RAG, LLM Agents, LangChain, Gemini API, Scikit-Learn, TensorFlow/Keras, PyTorch

PROFESSIONAL EXPERIENCE

Geospatial & Energy Data Engineering Intern 2026
Urban Pulse (Canadian startup), Remote / Tunisia

- Architected and containerized a multi-service geospatial data platform (Docker Compose, PostgreSQL, FastAPI, Streamlit) behind a business opportunity finder for retailers, franchisors and investors.
- Ingested and harmonized 37 municipal open datasets into PostgreSQL with schema validation and SQL transformations.
- Implemented SCD Type 2 pipelines to track infrastructure asset changes and keep 7 months of historical snapshots.
- Built FastAPI REST endpoints and an SLA monitoring pipeline measuring real-time ingestion latency.
- Developed spatial optimization (SciPy cKDTree) for energy and service-gap heatmaps and Dijkstra routing for infrastructure expansion.

AI Diagnostic & GenAI Systems Intern 2026
Sagemcom, Tunisia

- Built an AI diagnostic assistant for optical test benches (Gemini, Streamlit): hybrid RAG (error-code parsing plus TF-IDF) over 56 failures and a tool-calling agent generating maintenance tickets.
- Wrote a Pytest suite (16 unit tests, all passing) covering retrieval precision, agent responses and latency.

Machine Learning & Solar Energy Intern Jul - Aug 2025
TunBraille (startup), Tunisia

- Built an end-to-end solar power prediction pipeline (ESP32 sensors, Pandas, Scikit-Learn): Random Forest reached $R^2 = 0.94$ (MAE 9.5) versus 0.84 for the linear baseline; LSTM 0.95.
- Derived optimization recommendations (cleaning, panel tilt, battery scheduling) with an estimated 8 to 12% yield gain in simulation.

PROJECTS

Photovoltaic Time-Series Ingestion Engine (*Python, Pandas, Scikit-Learn*) 2025
Automated ETL and evaluation pipeline streaming weather and sensor feeds into solar yield models, with rolling-window features.

Pneumonia Detection from Chest X-rays (*TensorFlow/Keras, OpenCV | team of 2*) 2025
End-to-end pipeline on about 5,900 images: benchmarked Random Forest, XGBoost and CNNs (about 91 to 92% accuracy); fine-tuned EfficientNetB0 in a desktop app. github.com/hibames/pneumonia-detection

Autoencoder-based System Identification & MPC (*Python, Deep Learning | final-year project, team of 2*) 2026
Implemented the Partial Predictive Encoder (Automatica 2021): mean Best Fit Ratio 0.987 and 0.980 on two benchmarks; closed-loop MPC with stable tracking. github.com/hibames/Nonlinear-System-Identification-Using-Autoencoders-for-MPC

RAG Quiz Generator (*LangChain, Gemini API, Streamlit*) 2025
PDF-to-quiz pipeline with vector embeddings; tuned chunking and prompts to limit hallucinations. github.com/hibames/quiz-generator-ai

SOFT SKILLS, CSR (RSE) & HSE

Soft Skills: technical documentation, remote teamwork, autonomy, adaptability, rapid learning

CSR (RSE) & HSE: energy transition and green business focus, responsible AI, ESG awareness; safety-first, quality-oriented mindset (risk awareness, rigorous validation and testing)